

NISHANT SAI CHALLA

Victoria, BC

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Education

University of Victoria

Bachelor's of Computer Science and Mathematics

Summer 2026

Victoria, BC

Relevant Coursework

- Data Structures
- Software Design
- Algorithms Analysis
- Database Management
- ML and AI
- Data Mining
- Data Analysis
- Operating Systems

Experience

INSPIRE

Sept 2025 – Dec 2025

Software developer - Student

Victoria, BC

- Developing an educational platform that supports schools in teaching Indigenous languages and preserving culture.
- Redesigning legacy code into a modular structure, migrating CSS to Tailwind and MySQL to PostgreSQL.
- Designing and implementing gamified learning experiences and responsive home/feature pages across both the front and back end, and running QA testing to ensure platform reliability, usability, and readiness for classroom deployment.
- Managed work in Jira (sprint planning, ticket tracking), Slack and GitHub to coordinate delivery

University of Victoria

May 2025 – Aug 2025

AI/ML Student Researcher, Mathematics and Statistics Department

Victoria, BC

- Authored a research paper on Species Distribution Modelling using Deep Learning and Biotic Covariates.
- Developed a species distribution modelling pipeline to predict species presence/absence using Machine learning.
- Performed pre-processing, augmentation, pseudo-data generation, and model training using the StatEcoNet framework.
- Executed the project under the supervision of Arjun Banik (PhD Candidate) and Dr. Laura Cowen (Dean of Science), and currently preparing the manuscript for journal publication.

Projects

WEARLYZE | Python, YOLOv8-n, ResNet-50, MobileNet-V3 Large, ConvNeXt-Tiny, OpenAI CLIP, AWS

- Built an AI-powered fashion recognition app (WEARLYZE) that allows users to upload a photo to recognize clothing items and locate purchase options in SportChek.
- Pre-processed a dataset of 1M+ images with YOLOv8-n, fine-tuned models using ResNet-50, and deployed with MobileNet-V3 Large as part of a team project.
- Implemented similarity search on web-scraped SportChek data using SQLite and OpenAI CLIP, and evaluated model performance through testing with ConvNeXt-Tiny.

DERMADx | Python, PyTorch, EfficientNet-B4, AWS GPU Cloud and S3

- Developed an AI application to detect potential skin cancer from dermatoscopic images, supporting early diagnosis
- Trained on SIIM-ISIC 2018–2020, HAM10000, and PH2 datasets (Kaggle) and boosted minority-class recall through augmentation of positive cases.
- Trained models with the EfficientNet-B4 architecture on AWS g4dn.2xlarge GPU for scalable performance.

Spotify Streaming Data Analysis | R, SQL

- Analyzed Spotify's most popular songs and artists to find patterns in music consumption and streaming popularity.
- Applied linear and logistic regression to assess how artist streams predict top song performance and the likelihood of ranking in the top 10.
- Performed K-means clustering to segment artists by streaming behaviour and audience reach.

Technical Skills

Languages: C, Python, Java, JavaScript (ES6+), R, HTML5, CSS3, SQL

Frameworks & Libraries: React, RESTful APIs, Selenium, Scikit-learn, Pandas, OpenAI CLIP, TensorFlow, PyTorch

Tools & Platforms: SQLite, PostgreSQL, SSMS, Git, Linux, VS Code, DBeaver, Tableau, Azure, AWS

Hackathons/References/Clubs

Inspire2Dev Hackathon | Coding - Winner

NASA Space Apps 2025 | Participant

Arjun Banik | AI/ML Student Researcher Supervisor at UVic

Kezia Devathasan | INSPIRE Program Manager

VikeLabs | VikeLabs is a collective of students who learn to build, deploy, and test software apps.

SUBC | SUBC is a student team that designs, builds, and races a human-powered submarine in the UK and USA.